School of Artificial Intelligence and Big Data
Henan University of Technology
Zhengzhou 450001, China
lry@haut.edu.cn

August 5, 2026

The Editor-in-Chief
IEEE Transactions on Biomedical Engineering

Dear Editor,

We have prepared the following original research manuscript for consideration for publication in *IEEE Transactions on Biomedical Engineering*:

"FreeReg: Observability-Guided Non-Rigid Registration of Arbitrary 3D Models into Arbitrary Endoscopic Images and Videos"

Surgical mixed reality requires registering an arbitrary 3D model—a CT reconstruction, an instrument, an implant—into an arbitrary endoscopic image or video, computing the projection mapping among the model, camera, and pixel coordinate frames. Classical calibrate-then-align pipelines assume calibrated cameras, rigid targets, and independent per-frame estimation; all three fail in endoscopy. We formalize *free-coordinate registration* as the joint estimation of the free mapping $F = \pi_D \circ K \circ (R, t) \circ (I + W)$, with every component individually optional and degradable, and solve it with a multi-source correspondence engine with contour-gated arbitration, an observability-guided rigid-plus-warp estimator, a manifold SE(3) Kalman filter with innovation gating, a learned free-coordinate front-end refined by geometric ICP, and a co-anchored 3D Gaussian Splatting renderer.

Key results: (1) on the BOP Linemod benchmark the engine reaches BOP-protocol AR 0.848 versus FoundationPose 0.847, and on the full 13-object benchmark (2600 samples) both reach ADD-S 1.000—the Linemod ceiling; (2) on texture-less T-LESS the engine reaches ADD-S 0.942; (3) a learned front-end refined by ICP matches the geometric engine at 5× speed (ADD-S 0.987 at 158 ms); (4) the defining advantage appears in endoscopy, where general-purpose champions collapse out-of-distribution (FoundationPose TRE 104 mm, Endo3R ATE 34 mm) while our structural approach registers 12 clinical CT-to-bronchoscopy cases at 4.23 mm mean (92% <10 mm, Wilcoxon $p = 4.4 \times 10^{-31}$) at 30–80 FPS; (5) observability-guided non-rigid estimation recovers deformation to 1.1% on a real-colon FEM benchmark, and we isolate the rank-deficiency (151/648 directions), ray-sliding counterexample, and a determinant-corrupting parameterization pitfall in differentiable Lie-group optimization.

To our knowledge FreeReg is the only framework that is simultaneously rigid-capable, non-rigid, temporally coherent, occlusion-aware, real-time-renderable, and clinically deployable. The work aligns with IEEE TBME's scope in biomedical imaging methodology, surgical navigation, and computational modeling for mixed-reality guidance.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and declare no competing interests. The clinical data statement is provided in the accompanying Ethics Statement; the study is a retrospective secondary analysis of de-identified clinical imaging authorised through the institutional data-governance process of Henan Provincial People's Hospital.

Thank you for your consideration.

Sincerely,

Ranyang Li (on behalf of all authors)